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AN EXCELLENCE EQUATION: (E3 GOVERNANCE + 3E STRATEGY) ⇨ 3P PERFORMANCE

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Abstract

Purpose: Governance, sustainability, and enterprise excellence are united in an effort to promote continuously relevant and responsible organizations through Sustainable Enterprise Excellence (SEE), defined as:

- *SEE results as a consequence of balancing both the competing and complementary interests of key stakeholder segments, including society and the natural environment, to increase the likelihood of superior and sustainable competitive positioning and hence long-term enterprise success that is synonymous with continuously relevant and responsible governance, strategy, actions and results.*
- *This is accomplished through an integrated approach to organizational design and function emphasising innovation, operational, supply chain, customer-related, human capital, financial, marketplace, societal, and environmental performance. The intent of this approach is to ethically, efficiently and effectively (E3) integrate 3E (equity, ecology, economy) Triple Top Line strategy throughout enterprise governance, culture and activities to produce simultaneously pragmatic, innovative and sustainable Triple Bottom Line 3P (people, planet, profit) performance results.*

Methodology & Approach: Key elements and methods capable of assessing and advancing organizational progress toward SEE are identified, organized and developed.

Findings: Innovation, human capital, and foresight are key enablers of SEE. A SEE model called the Springboard together with Maturity Scales, graphical NEWS Compasses, and Narrative SWOT Plots that aid performance assessment and advancement are developed.

Implications for Further Research: Relationships of various SEE drivers to SEE performance are in need of further exploration wherein NEWS Compasses, Maturity Scales, and the Springboard may be further refined as SEE understanding advances.

Keywords: enterprise excellence, Innovation, enterprise excellence, maturity assessment, NEWS Compass, Springboard to SEE

Paper type: Conceptual paper

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1. Introduction

Drivers of organizational performance may derive purely from enterprise competitive needs or desires, or may be driven by technological progress or by legislative, ecological, or societal demands (Rao et al., 2000). Factors bordering on legion in number that serve as enablers of or barriers to enterprise performance exist and although many of these have been extensively explored, it is often the case than synergism among factors has the potential to exert the greatest leverage on performance.

Similarly, organizational forms and associated designs are influenced by both mega- and micro-trends. For example, over approximately the past century organizational forms have evolved that have been influenced first by the value for standardization associated with mass production then, later, adapted to address the demand for and value of customization. Increased globalization has more recently ratcheted up the competitive landscape across an array of business sectors and markets so that organizational forms morphed from prior forms or were created to extract the value inherent in strategically-and-demand-driven rapid innovation. Technology growth and capability have much to do the transition from standardization to customization to innovation and hence evolution in organizational forms. It is of course true that numerous enterprises combine standardization, customization, and innovation in their strategies and activities with forms that necessarily follow, that is: forms and functions are closely related.

The pace and scale of change indicates that megatrends and "mega-facts" may increase in both frequency and amplitude with some of these boding well and others of these boding poorly for humankind. In any case organizations must be cognizant of and responsive to such trends and facts.

In the mega-facts category we have that various sustainability-related factors such as human population, per capita material consumption, food consumption, water consumption, land use, urban sprawl, energy consumption, and carbon footprint have generally grown exponentially over a substantially extended period of time (Holdren and Ehrlich, 1974; Cohen, 2003). While some forecasting methods predict that human population growth will cease and possibly even reverse by the conclusion of the 21st century (Lutz et al., 2001), the cited sustainability factors have for a very long time been positively correlated so that, for example, decoupling of any of these can have profound impact. For example, technology innovation leading to breakthrough development of clean energy sources may not reduce energy consumption, but may transform the relative importance of energy consumption while also decreasing carbon footprint and boosting food production capability. We see in this single example the potential of human agency to influence the course of things.

With respect to megatrends it is clear that over the past three decades management strategy, policy, and practice have been shaped by crises of public

confidence, enterprise necessity for consistently high-level performance across an array of important dimensions, and increased pressure on organizations to contribute to both societal and environmental sustainability. These three respective forces are referred to as corporate governance, enterprise excellence, and corporate social and environmental responsibility. CSER is often simply referred to as sustainability, but with the implicit assumption that sustainability results follow responsible governance and actions.

Highly visible corporate scandals of the latter 1990s and early 2000s produced very clear and highly urgent demand for governmental action mandating transparency in corporate governance and thus contributed to establishment of the United Nations Global Compact in 2000 (Lawrence and Beamish, 2012) and, subsequently, to the Sarbanes-Oxley Act of 2002 (Lander, 2004; Svedin, 2012). In the context of enterprise excellence these influences manifest through formal assessment of enterprise leadership and governance. Governance is thus a non-trivial consideration that substantially impacts enterprise performance across numerous domains (Jackson, 1999; Wilkes, 2004), including financial performance (Erkens et al., 2012), knowledge (Martin-Castilla and Rodriguez-Ruiz, 2008), and CSER (McAdam and Leonard, 2003, Shahin and Zairi, 2007) as well as the overall market value of the firm (Black et.al., 2006).

Escalating globalization has greatly elevated competitive pressures that have in turn stimulated demand for and value of social and technological innovation and increased their importance in the design and delivery of products, processes, and services. The trend toward globalization has also highlighted the criticality of effective and efficient procurement and distribution of information and materials and significantly increased both the awareness and importance of efficient and effective supply chains.

Enterprise excellence uses complex business performance models such as those supporting influential international quality awards and the balanced scorecard (Kaplan and Norton, 1992). In comparison, sustainability concerns gave birth to such widely used standards as the ISO 14000 Environmental Management Standard (Castka and Balzarova, 2008) and the ISO 26000 Corporate Social Responsibility Standard (Schwartz and Tilling, 2009) along with such responsibility-oriented approaches as the Global Reporting Initiative (Eccles and Krzus 2010). Although sustainability has many facets, including financial ones, as regarded by these standards and approaches sustainability is for practical purposes often reduced to societal and environmental dimensions.

Described as previously, sustainability is well captured by the phrase: “lean, green, ethical and real” (Edgeman and Eskildsen, 2012). The phase implies enterprise practices that exercise great care in resource consumption or all sorts, formulation of and adherence to socially and environmentally constructive policy and practice, transparency of conduct, and delivery of “lean and green” results.

"Lean, green, ethical and real" sustainability has been identified as an emerging megatrend (Lubin and Esty, 2010) and an emerging source of competitive advantage (Laszlo and Zhexembayeva, 2011), wherein effective enterprise sustainability policy – especially environmental policy – is a documented contributor to firm value (Al-Najjar and Anfimiadou, 2012). Similarly, effective and efficient implementation of enterprise excellence approaches emphasizing performance across an array of key domains has also proven to contribute to firm value (Balasubramanian et al., 2005).

Focused through an enterprise lens we may summarize by stating that enterprise governance should be effective, efficient and ethically oriented (E3). Organizational leadership is then charged with the responsibility of formulating and deploying strategy that emphasizes equitable (humane) commitment of enterprise resources, ecological stewardship, and economic responsibility that will be referred to as 3E Triple Top Line (TTL) strategy (McDonough and Braungart, 2002a). Superior connection of E3 governance and 3E strategy in turn deliver positive performance results with respect to society (people), the natural environment (planet), and financial areas (profit) or 3P Triple Bottom Line (TBL) performance (Elkington, 1997).

An easily recalled expression representing these ideas may be referred to as the "*Sustainable Enterprise Excellence (SEE) Equation*":

(E3 Governance + 3E Strategy) ⇨ 3P Performance

2. Sustainable Enterprise Excellence and Sustainability

Sustainable Enterprise Excellence or SEE integrates corporate governance, sustainability and enterprise excellence and the SEE Equation represents one possible quasi-mathematical expression of the continuously relevant and responsible organization (CR2O) envisioned by SEE, wherein:

SEE results as a consequence of balancing both the competing and complementary interests of key stakeholder segments, including society and the natural environment, to increase the likelihood of superior and sustainable competitive positioning and hence long-term enterprise success that is synonymous with continuously relevant and responsible governance, strategy, actions and results.

This is accomplished through an integrated approach to organizational design and function emphasising innovation, operational, supply chain, customer-related, human capital, financial, marketplace, societal, and environmental performance. The intent of this approach is to ethically, efficiently and effectively (E3) integrate 3E (equity, ecology, economy) Triple Top Line strategy throughout enterprise governance, culture and activities to produce simultaneously pragmatic, innovative and sustainable Triple Bottom Line 3P (people, planet, profit) performance results.

Sustainability is foundational to SEE so that it is important to delineate how its construction in this context.

Sustainability is in most contexts regarded as a 'capacity to endure'. Further, we have already insinuated that it should be "lean, green, ethical and real". Connecting sustainability to excellence and governance, however, alters sustainability from what may be described as capacity to maintain *status quo* or move marginally forward, toward a stronger construct. That is SEE regards sustainability as follows.

Sustainability is the propensity to improve in both absolute terms and in relative terms as driven by enterprise competitive context.

With respect to a given enterprise then, sustainability should manifest as economic viability and positive contribution to both societal and environmental sustainability. Exceptional enterprises should excel within this context, means for which are later suggested herein. Such sustainability through E3 governance, 3E strategy, and 3P performance is, however, ever more complex.

3. Sustainable Enterprise Excellence and Innovation

Reality most typically dictates primacy of the profit performance domain since an enterprise that is not sufficiently economically secure will ordinarily contribute inadequately to society and will generate similarly inadequate environmental performance. Despite primacy of the financial domain, societal concerns, consumer demands, stakeholder expectations, and regulatory compliance have fueled significant urgency for sustainable performance in all 3P areas. Thus far however, rapidly increasing societal and environmental challenges have outpaced organizational delivery capability and capacity with respect to such performance.

Capability and capacity are multi-faceted, with one aspect being enterprise intelligence concerning societal and environmental challenges and a second aspect being the ability to embed innovation of the needed scope, scale, trajectory, and velocity in enterprise strategy, policy, and practice.

Innovation has been cited as a key enabler of sustainability (Nidumolu et al., 2009) and it is this aspect of innovation that is here of central importance as a driver of sustainable enterprise excellence.

In particular, socio-ecological innovation (SEI) weds sustainable innovation and innovation for sustainability and is related to the organizational innovation capacity factor – a factor that may be analysed through the innovation lens of the balanced scorecard in its original form. Sustainable innovation is an element of an enterprise's cultural fabric wherein innovation is regular, rigorous, systematic and systemic to enterprise strategy and practice (Skarzynski and Gibson, 2008). In contrast, innovation for sustainability is innovation that specifically targets

societal or environmental impact and contribution (Cooperrider, 2008). Solely environmentally focused innovation is commonly referred to as eco-innovation (Carillo-Hermosilla et al., 2009) whereas innovation that mirrors designs and strategies found in nature is referred to as biomimicry (Benyus, 2002). In general SEI is consistent with cradle-to-cradle innovation and design philosophy (McDonough and Braungart, 2002b).

Large-scale deployment of SEI throughout both an enterprise and its ecosystem is possible through a modified form of quality function deployment (Xie et al., 2003) developed by Edgeman and Hensler (2005). Socio-ecological innovation in general and extending throughout an enterprise ecosystem in particular is important with respect to enterprise progress toward SEE and extension of across multiple and connected enterprise ecosystems contributes to creation of sustainable enterprise economies (Waddock and McIntosh, 2011). Seen from a societal perspective SEI is of broader importance since it contributes to large scale *socio-ecological resilience* (Olsson and Galaz, 2011) where resilience may be regarded as the capacity of a system to confront challenges and change, yet continue to positively develop.

SEI is central to SEE and hence in any reasonable model thereof. Edgeman and Eskildsen (2012) provide a blueprint for embedding socio-ecological innovation in enterprise culture and have in related work addressed SEI maturity assessment.

Although not emphasized in SEE the emerged reality of "big data" demands consideration – not only with respect to enterprise excellence, sustainability, and governance, but much more pervasively in response to rapidly escalating need to make sense of massive amounts of information in order to more responsibly formulate and operationalize strategy and deliver results. Big data has particular potential for influencing both targets of innovation and the rapidity with which innovation occurs. Significant energy is being devoted to derivation of useful "big data analytics" (Franks, 2012) and big data competitive strategy (Davenport and Harris, 2007). The reality of "big data" is driven not only by the obvious influences of globalization and technology growth, but by human knowledge expansion that has grown exponentially as a knowledge-based generalization of Moore's Law (Schaller, 1997).

In particular SEI is regarded as a key means of activating 3E TTL strategy to yield 3P TBL performance and as such is regarded as a primary driver of SEE. This implies that SEI in focal to any SEE-oriented business model, organizational form, and organizational design.

This latter statement is made with cognizance that to the extent that SEE is attained, it is within the context of organizational form, hence making organizational design central to SEE and is embedded in the formulation of SEE. Although not addressed in any substantial detail herein, organizational design is a formal, guided process for integrating the human capital, information and

technology of an organization. Organizational design is used to match the form of the enterprise as closely as possible to enterprise purpose(s)– SEE in the present context. It is through the organizational design process that enterprises act to improve the likelihood that its collective efforts will succeed.

4. Modelling Sustainable Enterprise Excellence: the Springboard to SEE & Assessment

Many factors critical to SEE and hence to optimization of its equation are treated seriously by the balanced scorecard or by enterprise excellence models. Other key factors are addressed by, for example, the Global Reporting Initiative or in the expectations of annual Communication of Progress reports required of United Nations Global Compact members. Governance is either explicitly addressed or is insinuated by all of these models and approaches.

Effective integration of corporate governance, sustainability, and enterprise excellence has, however, proven more challenging. While various efforts to integrate these into a single model have been attempted, with two primary approaches having received a majority of the attention, all models thus far developed have delivered only limited success.

Models of a first kind add a sustainability module to an excellence model (Asif et al., 2011), whereas models of a second kind attempt to successfully incorporate financial performance in CSER oriented models (Zwetsloot, 2003). In contrast a model of a third kind aims at integration of governance, sustainability and enterprise excellence from the outset with modified forms of the balanced scorecard providing one sort of example (Zingales and Hockerts, 2003) and a conceptual model referred to as 3C-SR (Meehan et al., 2006) providing another.

Models generally emphasize and assess that which they value: enterprise excellence (1E strategy ⇨ 1P performance) or sustainability (2E strategy ⇨ 2P performance), whereas SEE values a transfer of the form (E3 governance + 3E strategy) ⇨ 3P performance. SEE is thus "third kind" in orientation so that the *Springboard to SEE* is a new model of the third kind.

The *Springboard* explicitly integrates sustainability, excellence and governance efficiently, effectively, and ethically from strategy through results and back to strategy, that is, as a continuous cycle of excellence. Specifically the *Springboard* aims both to model SEE and assess the progress of enterprise efforts to translate the 3E (equity, ecology, economy) Triple Top Line strategy into 3P (people, planet, profit) Triple Bottom Line through E3 governance. In acknowledging the importance of (E3 governance + 3E strategy) ⇨ 3P results transference it must be recognized that transference occurs not as three one-to-one E ⇨ P relationships of the form equity strategy ⇨ people results, ecological strategy ⇨ planet results, and economic strategy ⇨ profit results. Instead all

three (E3 + 3E) components are integrated with *each* of equity, ecology, and economy delivering results in *all* three TBL performance domains, as portrayed in Figure 1.

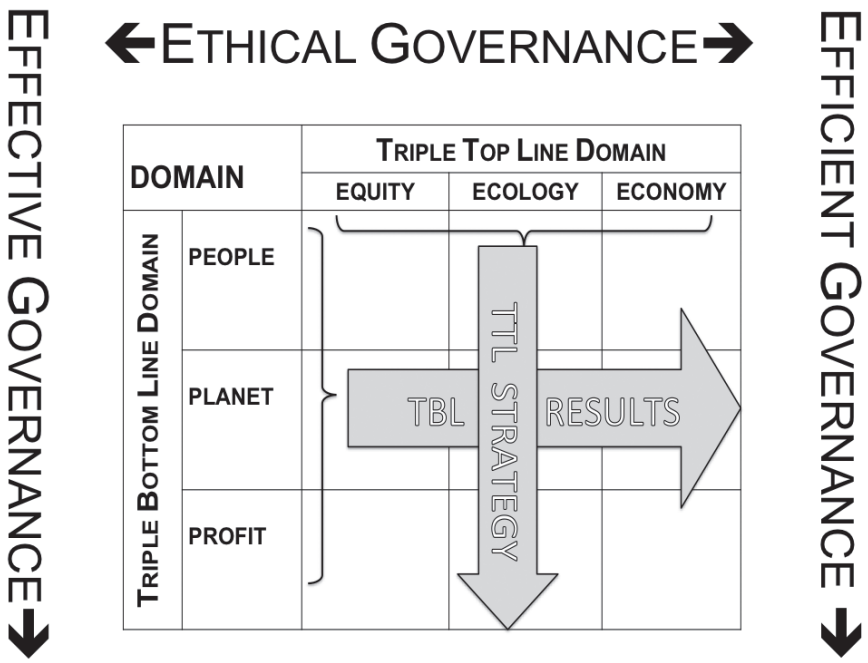

Figure 1. Equity ⇨ 3P results, Ecology ⇨ 3P results, Economy ⇨ 3P results

As its object the *Springboard* of Figure 2 intends to actualize the SEE equation and hence progress toward SEE. More explicitly, equity ⇨ people, ecology ⇨ planet, and economy ⇨ profit synapses represent creation of enhanced sustainable societal performance, sustainable environmental performance, and sustainable financial performance. SEE connects the kernels of governance, sustainability and enterprise excellence as a means of delivering responsible competitiveness (Avlonas and Swannick, 2009).

From left-to-right the *Springboard* is applied internal to an organization's form and design. The model itself is divided into three primary blocks, beginning with E3 governance and 3E strategy, the agents of which are policies, people and partnerships. Fundamentally however, it is in the process implementation and execution block where the SEE equation is executed, that is, where (E3 governance + 3E strategy) are transformed into 3P performance that is reflected in the rightmost model results and refinement block wherein performance is divided into the four primary areas of human capital results, innovation results, financial results, and sustainability results. As the enterprise advances, new intelligence

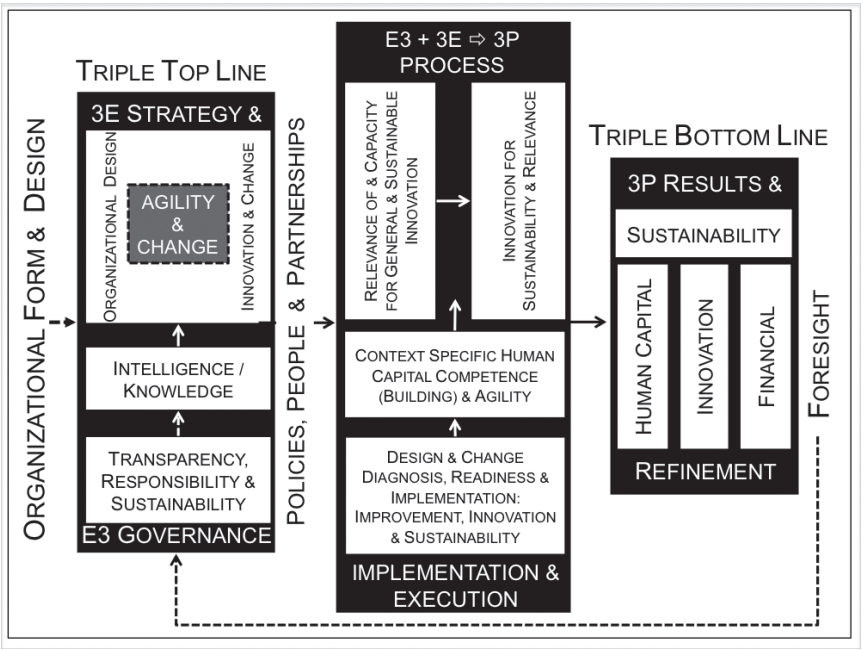

Figure 2.
Springboard to SEE model

and foresight critical organizational performance is generated that informs the next cycle.

The *Springboard* regularly (most typically an annual cycle), rigorously, and systematically evaluates all key organizational strategies, activities and results. Assessment is intended to provide performance feedback, suggest necessary changes, and point the organization toward both current and *next* best practices and sources of competitive advantage through learning and foresight that drive relevant and responsible actions and results that are consistent with the objectives of *Sustainable Enterprise Excellence*. *Springboard* assessment yields an overall *Springboard to SEE NEWS Report* formed by coalescing the results of six (graphical) *NEWS Compasses* and companion *SWOT Plot Narratives* into an integrated SEE performance dashboard (Eckerson, 2006).

One *NEWS Compass – SWOT Plot Narrative* pair is based on the strategy and governance block of the *Springboard*, another pair on the process and performance implementation and execution block, and the four remaining pairs are in one-to-one correspondence with the human capital, innovation, financial, and sustainability performance categories of the *Springboard* results and refinement block. Figures 3 and 4 portray a generic *NEWS Compass* and generic *SWOT Plot Narrative*, respectively. SWOT analysis focuses on enterprise strengths, weaknesses, opportunities and threats relative to specifically selected aspects of SEE performance with strengths and weaknesses reflecting enterprise

internalities and opportunities and threats reflecting externalities (Heuer and Pherson, 2011).

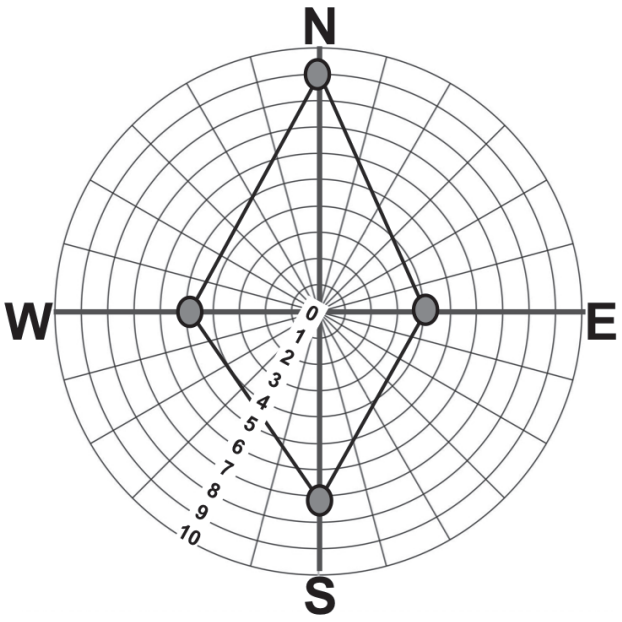

Figure 3.
Springboard NEWS Compass

NEWS Compasses are modified radar charts (Yau, 2011) that graphically summarize assessment results. *NEWS Compasses* such as the generic one presented in Figure 3 are constructed by assessing level of fulfilment or maturity for each of four defined criterion areas, called compass points due to the appearance of the figure, the intent of the assessment to provide information (or "news") regarding the current state of the enterprise, and direction toward next best practices and sources of competitive advantage, hence N-E-W-S as a mnemonic that might also be associated with "north-east-west-south".

Maturity evaluation of each N-E-W-S area is based on specifically derived 0-to-10 point scales that may be obtained from the authors. *NEWS Compass* points for each of the six compasses are cited in Table 1 and are derived from comprehensive consideration of multiple sources of sustainability, enterprise excellence, innovation, governance and associated criteria where sources include but are not limited to the philosophy or models, principles and criteria found in documentation for America's Baldrige National Quality Award, the European Quality Award, the Global Reporting Initiative, and the Ten Principles of the United Nations Global Compact. Baldrige and European Quality Award documentation are particularly instructive with respect to maturity assessment of such criteria.

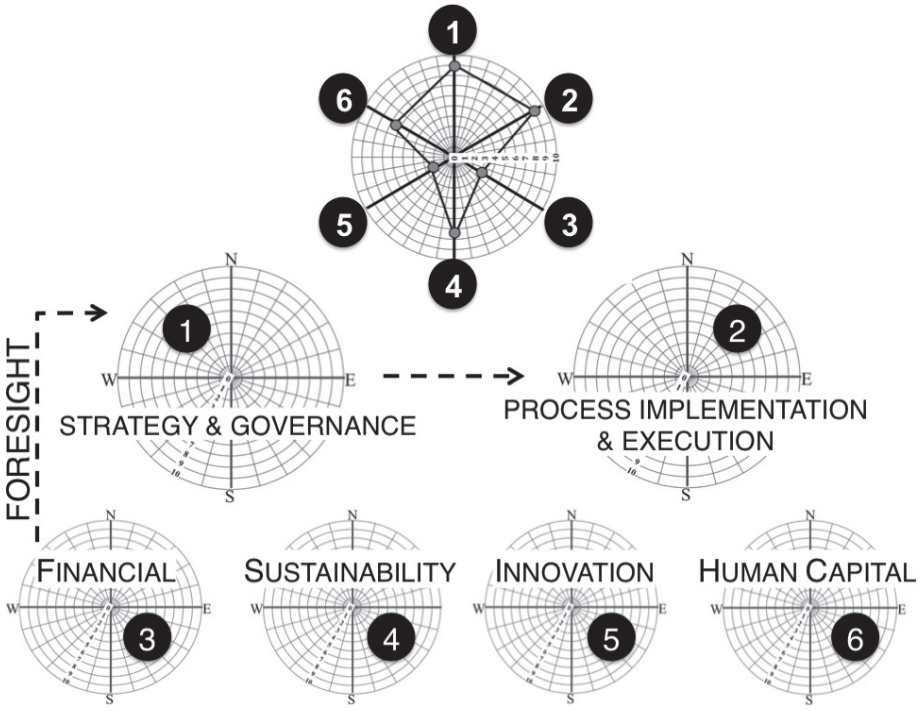

Figure 4.
Springboard SWOT Plot

COMPASS POINT	SPRINGBOARD ASPECTS & COMPASS POINTS
ASPECT	STRATEGY & GOVERNANCE
N	Transparency, Responsibility and Sustainability
E	Intelligence & Knowledge Building & Acquisition
W	Organizational Design: Human Capital
S	Organizational Design: Innovation & Change
ASPECT	PROCESS IMPLEMENTATION & EXECUTION
N	Relevance of & Capacity for Specific & Sustainable Innovation
E	Design & Change Diagnosis, Readiness & Implementation
W	Context Specific Human Capital Competence Building & Agility
S	Innovation for Sustainability & Relevance
ASPECT	FINANCIAL PERFORMANCE RESULTS
N	Sustainability Investment & Return
E	Supply / Value Chain & Other Performance Improvement
W	Human Capital Investment & Return
S	R&D: Return on & Reinvestment in Innovation
ASPECT	SUSTAINABILITY PERFORMANCE RESULTS

Table 1.
Springboard NEWS Compass Points

COMPASS POINT	SPRINGBOARD ASPECTS & COMPASS POINTS
N	Financial Results & Refinement Associated with Sustainability Efforts
E	Societal Sustainability Results & Refinement
W	Human Capital Results & Refinement Associated with Sustainability
S	Environmental Sustainability Results & Refinement
ASPECT	INNOVATION PERFORMANCE RESULTS
N	Innovation for Sustainability: Society and the Environment
E	Other Context Specific Innovation
W	Business Model Innovation
S	Sustainable Innovation
ASPECT	HUMAN CAPITAL PERFORMANCE RESULTS
N	Innovation Capacity & Performance
E	Specific & General Competence & Agility
W	Strategic & Tactical Continuous Improvement
S	Sustainability Intelligence & Performance: Society and the Environment

Table 1.
Continued

5. Comments on Assessment

The *Springboard*, its *NEWS Compasses*, *SWOT Plot Narratives* and the overall *NEWS Report* are intended to form a modelling and assessment system simple enough to be used by organizations novice in their excellence and sustainability experience and expertise, yet useful to enterprises that are sophisticated. Each *SWOT Plot Narrative* should be integrated and aligned with its corresponding *NEWS Compass* assessment. This summary narrative must be thoughtfully formed due to its narrative nature and requires the assessment team to regard importance relative to enterprise context of the various assessed areas represented by the six primary compasses and narratives.

Given the nature of SEE, themes such as governance, sustainability, innovation, intelligence and foresight generation, competence building, and organizational design should be emphasized and aligned throughout the assessment process. In other words, during the assessment it is critical to be aware of what is, and is not being assessed.

As a composition of the six primary compasses, a *Summary NEWS Compass* (Figure 5) will have six axes and should be aligned with a *Summary SWOT Plot Narrative* (Figure 6) that, similar to the *Summary NEWS Compass*, is formed as a composite of the six SWOT Plot Narratives associated with the primary compasses. The *NEWS Compass Dashboard* and *SWOT Plot Narrative Dashboard* feed the *Summary NEWS Compass* and *Summary SWOT Plot Narrative* to yield the overall *Springboard NEWS Report*.

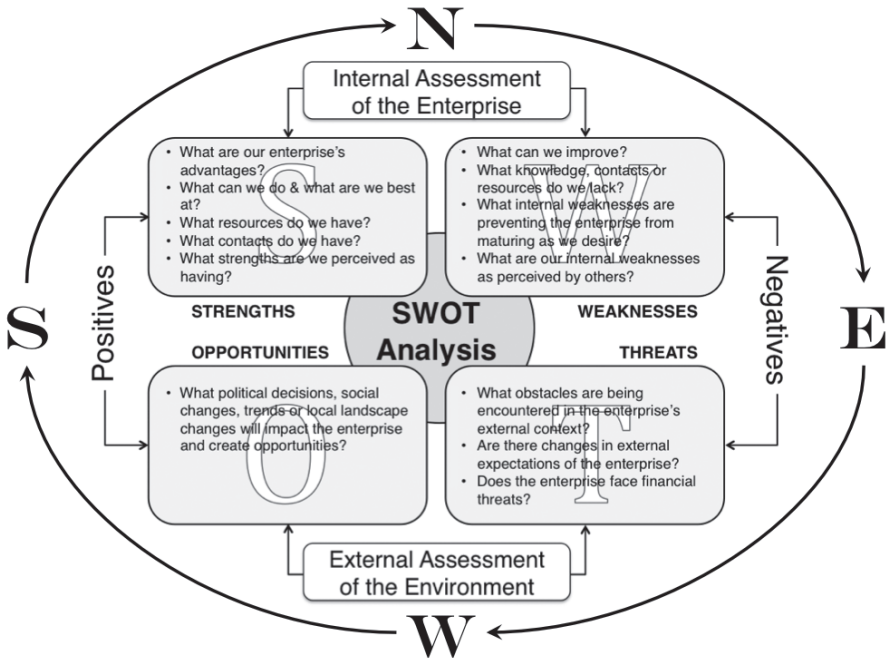

Figure 5.
NEWS Compass Dashboard

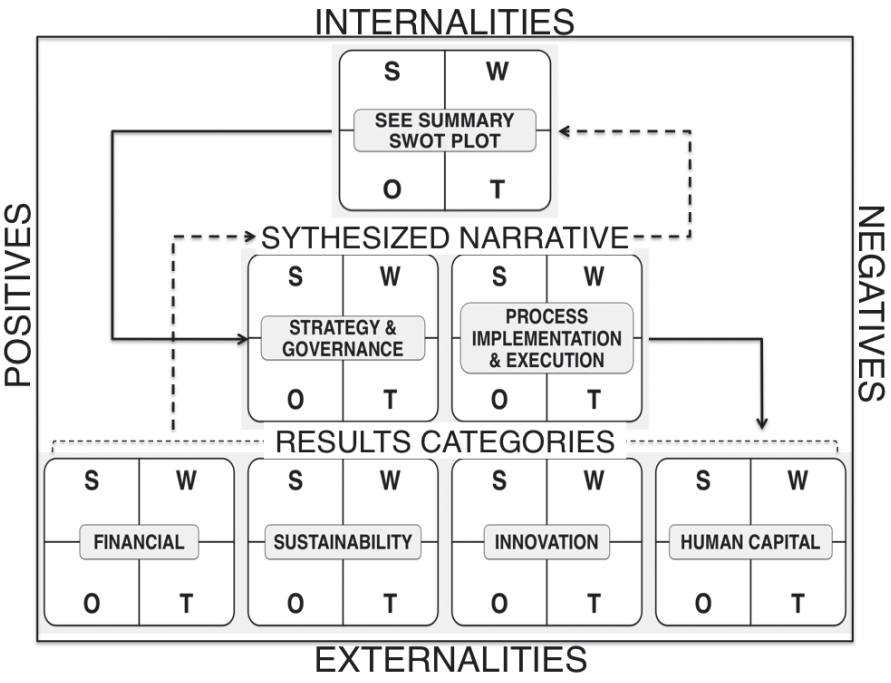

Figure 6.
SWOT Plot NEWS Narratives Dashboard

The *Springboard* neither over- nor undervalues elements relative to one another since any weighting of elements is left to the enterprise. Whether mathematically or descriptively, valuation for a given axis of any compass and, indeed, the compasses themselves should be determined according to enterprise context since it is unlikely that the various axes are of equal importance within a compass, let alone from compass to compass – as with all other things one must be constantly mindful of context.

Customers are central to excellence systems and although customers are nowhere explicitly cited in Table 1, they are key to the *Springboard*, with their importance underscored by the word *relevance* appearing in the table, inherent in all references to innovation for sustainability, and implied in all societal dimensions.

SEE and the *Springboard* aim to aid organizations in the quest to become continuously relevant and responsible by driving E3 (efficient, effective and ethical) governance and 3E (equity, ecological and economic) strategy to produce 3P (people, planet and profit) enterprise results. This requires a cycle of assessment, generation and implementation of usable foresight leading to next best practices and sources of competitive advantage.

6. Summary

Sustainable Enterprise Excellence (SEE) has been defined in order to underscore the important roles organizations play in contributing to sustaining both society and the natural environment, while also being themselves sustainable. In particular SEE is approached through intentional combination of ethical, effective and efficient (E3) governance with societally equitable, ecologically sound, and economically viable (3E) strategy aimed at producing superior (e.g. excellent) sustainable results in three primary performance domains: people (society), planet (environment), and profit (financial) (3P). As such there is a conceptual (rather than mathematical) equation associated with SEE: E3 Governance + 3E Strategy ⇨ 3P Performance.

As with other excellence constructs SEE is facilitated and advanced by assessment of enterprise performance relative to an underlying model and meaningful associated criteria. Model criteria should be evaluated based on a combination of both lagging and leading indicators that reflect current organizational performance, forecast future performance needs and levels, and suggest next best practices and sources of competitive advantage.

In addition to being useful, assessment should be of sufficient ease so as to actually be used. This combination of useful and used suggests that SEE might be encouraged by a relatively simple model and highly consumable assessment format. Toward this end a simple SEE model referred to as the Springboard was introduced and a straight-forward assessment approach based on a combined graphical and narrative approach was developed.

The intent of the present work is that, taken together, the SEE construct, model, and assessment regime should aid enterprises interested in positively contributing not only to their own continuity and economic security, but toward a healthier society that functions in a healthier natural environment.

References

Al-Najjar, B., Anfimiadou, A. (2012), "Environmental policies and firm value", *Business Strategy and the Environment*, Vol. 21, No. 2, pp. 49–59.

Asif, M., Searcy, C., Garvare, R., Ahmad, N. (2011), "Including sustainability in business excellence models", *Total Quality Management & Business Excellence*, Vol. 22, No. 7, pp. 773–786.

Avlonas, N., Swannick, J. (2009), "Developing Business Excellence While Delivering Responsible Competitiveness", in: Eskildsen, J., Jonker, J. (Eds.), *Management Models for the Future*, Springer, Berlin, pp. 171–184.

Balasubramanian, S., Mathur, I., Thakur, R. (2005), "The impact of high-quality firm achievements on shareholder value: focus on Malcolm Baldrige and J.D. Power and Associates awards", *Journal of the Academy of Marketing Science*, Vol. 33, No. 4, pp. 413–422.

Benyus, J. (2002), *Biomimicry: Innovation Inspired by Nature*, HarperCollins, New York.

Black, B., Jang, H., Kim, W. (2006), "Does corporate governance predict firms' market values? Evidence from Korea", *The Journal of Law, Economics & Organization*, Vol. 22, No. 2, pp. 366–413.

Carillo-Hermosilla, J., del Rio Gonzalez, P., Konnola, T. (2009), *Eco-Innovation: When Sustainability and Competitiveness Shake Hands*, Palgrave Macmillan, New York.

Castka, P., Balzarova, M. (2008), "The impact of ISO 9000 and ISO 14000 on standardisation of social responsibility: an inside perspective", *International Journal of Production Economics*, Vol. 113, No. 1, pp. 74–87.

Cohen, J. (2003), "Human population: the next half century", *Science*, Vol. 302, pp. 1172–1175.

Cooperrider, D. (2008), "Sustainable innovation", *BizEd*, July-August, pp. 32–38.

Davenport, T., Harris, J. (2010), *Competing on Analytics: The New Science of Winning*, Harvard Business Press, Boston.

Eccles, R. and Krzus, M. (2010), *One Report: Integrated Reporting for a Sustainable Strategy*, John Wiley, Hoboken, NJ.

Eckerson, W. (2006), *Performance Dashboards: Measuring, Monitoring, and Managing Your Business*, John Wiley, New York.

Edgeman, R., Eskildsen, J. (2012), "Viral innovation: integration via sustainability and enterprise excellence", *Journal of Innovation and Best Business Practice*, Vol. 2012, 13 pp., DOI: 10.5171/2012.361451.

Edgeman, R., Hensler, D. (2005), "QFD and the BEST paradigm: deploying sustainable solutions", *World Review of Science, Technology and Sustainable Development*, Vol. 2, No. 1, pp. 49–59.

Elkington, J. (1997), *Cannibals with Forks: The Triple Bottom Line of 21st Century Business*, Capstone Publishing, Oxford, UK.

Erkens, D., Hung, M., Matos, P. (2012), "Corporate governance in the 2007–2008 financial crisis: evidence from financial institutions worldwide", *Journal of Corporate Finance*, Vol. 18, No. 2, pp. 389–411.

Franks, B. (2012), *Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics*, John Wiley & Sons, Hoboken, NJ.

Heuer, R., Pherson R. (2011), *Structured Analytic Techniques for Intelligence Analysis*, CQ Press, Washington, DC.

Holdren, J., Ehrlich, P. (1974), "Human population and the global environment", *American Scientist*, Vol. 62, No. 3, pp. 282–292.

Jackson, S. (1999), "Exploring the possible reasons why the UK government commended the EFQM excellence model as the framework for delivering governance in the new NHS", *International Journal of Health Care Quality Assurance*, Vol. 12, No. 6, pp. 244–253.

Kaplan, R.S., Norton, D.P. (1992), "The balanced scorecard: measures that drive performance", *Harvard Business Review*, January-February, pp. 71–79.

Lander, G. (2004), *What is Sarbanes-Oxley?*, McGraw Hill, New York.

Lawrence, J., Beamish, P. (Eds.) (2012), *Globally Responsible Leadership: Managing According to the UN Global Compact*, Sage, Los Angeles.

Laszlo, C., Zhexembayeva, N. (2011), *Embedded Sustainability: the Next Big Competitive Advantage*, Greenleaf Publishing, Sheffield, UK.

Lubin, D., Esty, D. (2010), "The sustainability imperative", *Harvard Business Review*, May, pp. 2–10.

Lutz, W., Sanderson, W., Scherbov, S. (2001), "The end of world population growth", *Nature*, Vol. 412, August, pp. 543–545.

Martin-Castilla, J., Rodriguez-Ruiz, O. (2008), "EFQM model: knowledge governance and competitive advantage", *Journal of Intellectual Capital*, Vol. 9, No. 1, pp. 133–156.

McAdam, R., Leonard, D. (2003), "Corporate social responsibility in a total quality management context: opportunities for sustainable growth", *Corporate Governance*, Vol. 3, No. 4, pp. 36–45.

McDonough, W., Braungart, M. (2002a), "Design for the triple top line: new tools for sustainable commerce", *Corporate Environmental Strategy*, Vol. 9, No. 6, pp. 251–258.

McDonough, W., Braungart, M. (2002b), *Cradle to Cradle: Remaking the Way We Make Things*. North Point Press, New York.

Meehan, J., Meehan, K., Richards, A. (2006), "Corporate social responsibility: the 3C-SR model", *International Journal of Social Economics*, Vol. 33, Nos. 5–6, pp. 386–398.

Nidumolu, R., Prahalad, C.K., Rangaswami, M.R. (2009), "Why sustainability is now the key driver of innovation", *Harvard Business Review*, *September*, pp. 57–64.

Olsson, P., Galaz, V. (2011), "Social-Ecological Innovation and Transformation", in: Nicholls, A., Murdoch, A. (Eds.), *Social innovation: Blurring Boundaries to Reconfigure Markets*, Palgrave MacMillan, London, pp. 223–247.

Rao, H., Morrill, C., Zald, M. (2000), "Power Plays: How Social Movements and Collective Action Create New Organizational Forms", in: Shaw, B., Sutton, R. (Eds.), *Research in Organizational Behaviour*, Vol. 22, Elsevier Science, New York, pp. 237–281.

Schaller, R. (1997), "Moore's law: past, present and future", *IEEE Spectrum*, Vol. 34, No. 6, pp. 52–59.

Schwartz, B., Tilling, K. (2009), "ISO-lating corporate social responsibility in the organizational context: a dissenting interpretation of ISO 26000", *Corporate Social Responsibility and Environmental Management*, Vol. 16, No. 3, pp. 289–299.

Shahin, A., Zairi, M (2007), "Corporate governance as a critical element for driving excellence in corporate social responsibility", *International Journal of Quality & Reliability Management*, Vol. 24, No. 7, pp. 753–770.

Skarzynski, P., Gibson, R. (2008), *Innovation to the Core: a Blueprint for Transforming the Way Your Company Innovates*, Harvard Business Press, Boston.

Svedin, L. (2012), *Accountability in Crises and Public Trust in Governing Institutions*, Routledge, New York.

Waddock, S., McIntosh, M. (2011), *SEE Change: Making the Transition to a Sustainable Enterprise Economy*, Greenleaf Publishing, Sheffield, UK.

Wilkes, J. (2004), "Corporate governance and measuring performance", *Measuring Business Excellence*, Vol. 8, No. 4, pp. 13–16.

Xie, M., Tan, K.C., Goh, T.N. (2003), *Advanced QFD Applications*, ASQ Quality Press, Milwaukee, WI.

Yau, N. (2011), *Visualize This: The Flowing Data Guide to Design, Visualization, and Statistics*, John Wiley, New York.

Zingales, F., Hockerts, K. (2003), "Balanced scorecard and sustainability: examples from literature and practice", Center for the Management of Environmental Responsibility Working Paper Series 2003/30/CMER, INSEAD, Fontainebleau Cedex, France, available at: http://flora.insead.edu/fichiersti_wp/inseadwp2003/2003–30.pdf (accessed 10 November 2012).

Zwetsloot, G. (2003), "From management systems to corporate social responsibility", *Journal of Business Ethics*, Vol. 44, Nos. 2–3, pp. 201–208.